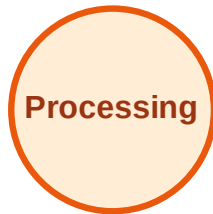


BFS Traversal

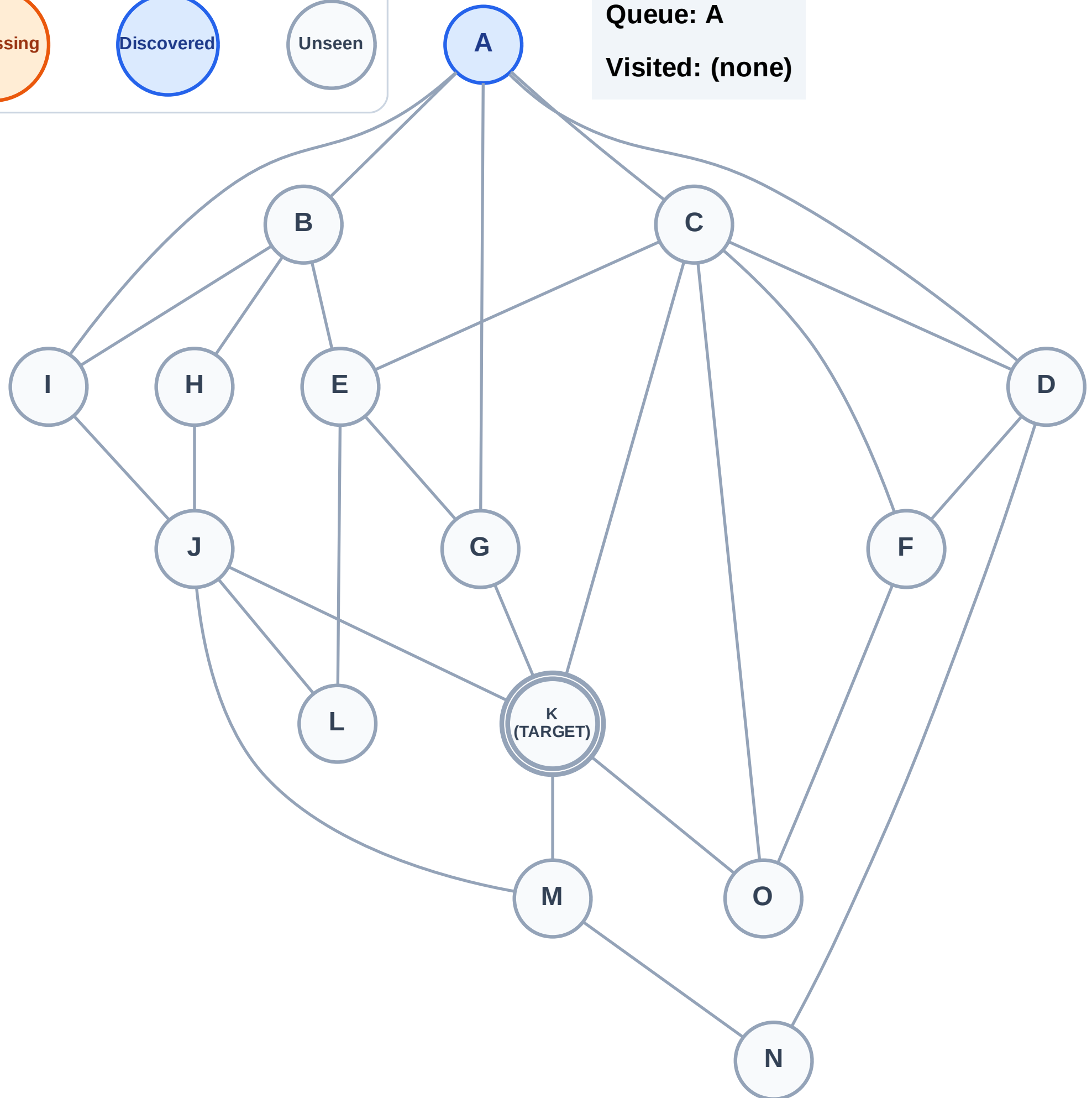
Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Queue: A

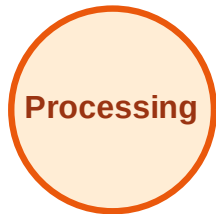
Visited: (none)



BFS Traversal

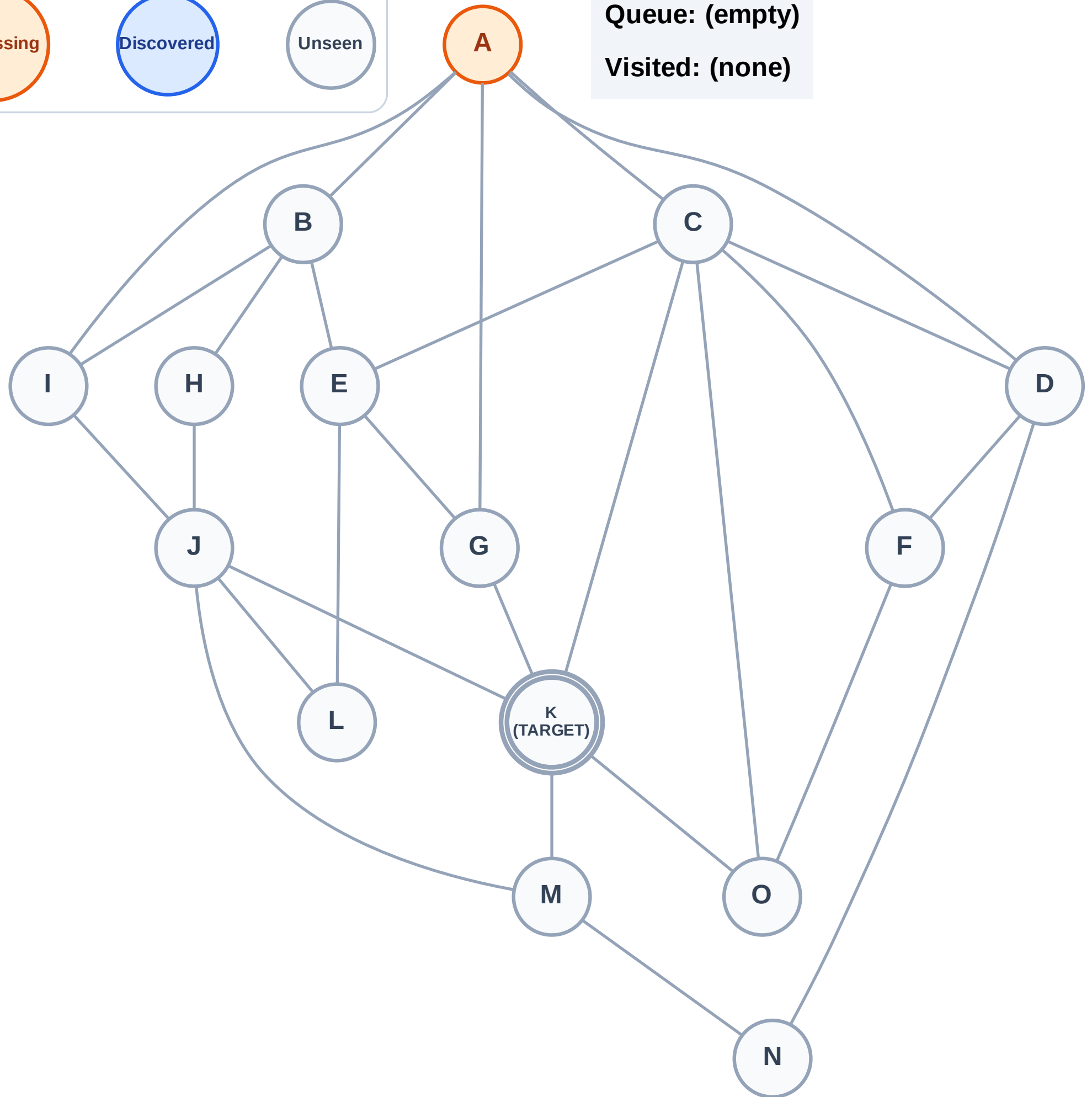
Step 2: Process node A

Legend



Queue: (empty)

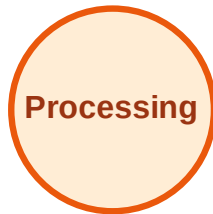
Visited: (none)



BFS Traversal

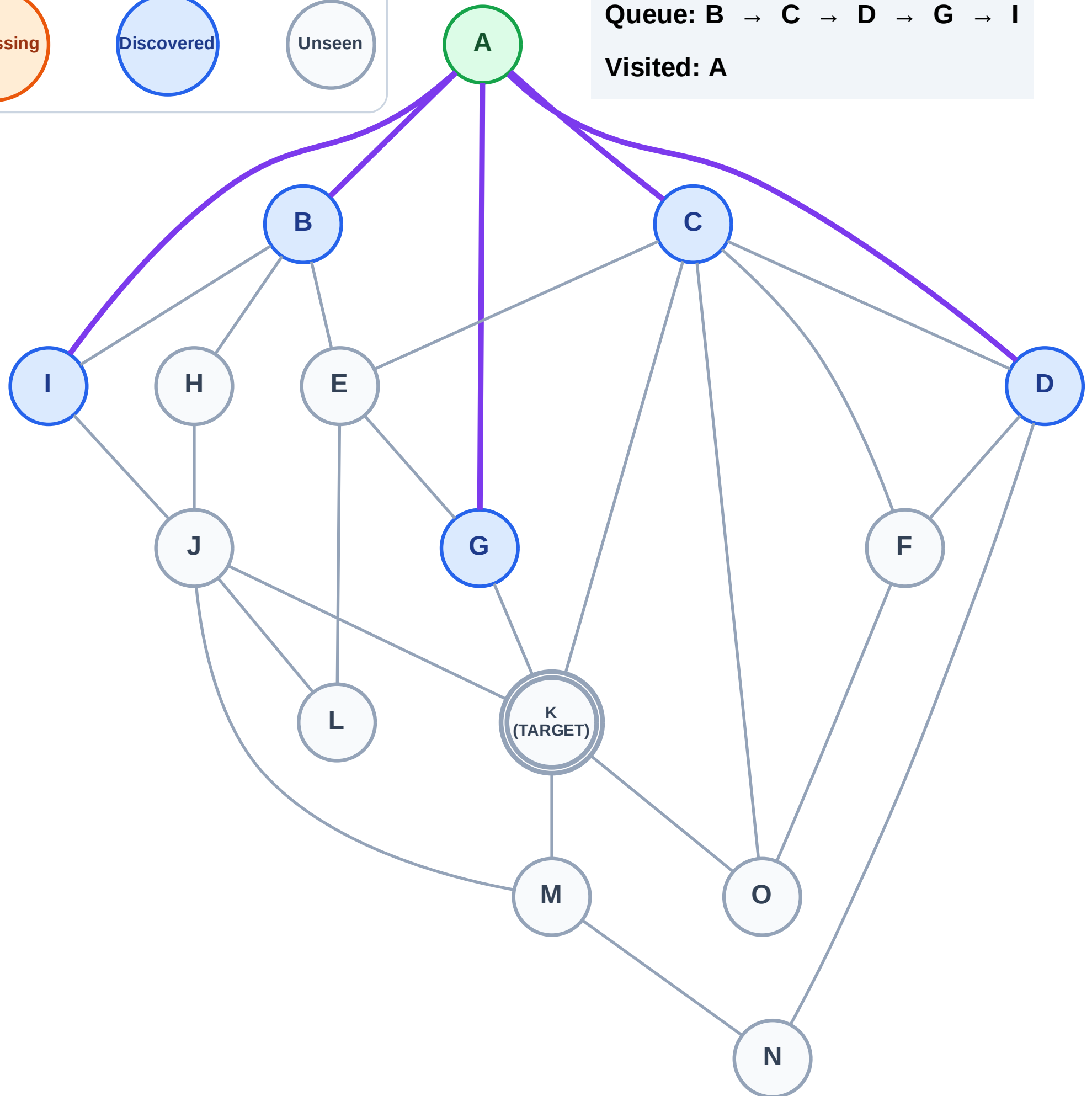
Step 3: Discover B, C, D, G, I from A

Legend



Queue: B → C → D → G → I

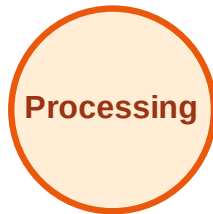
Visited: A



BFS Traversal

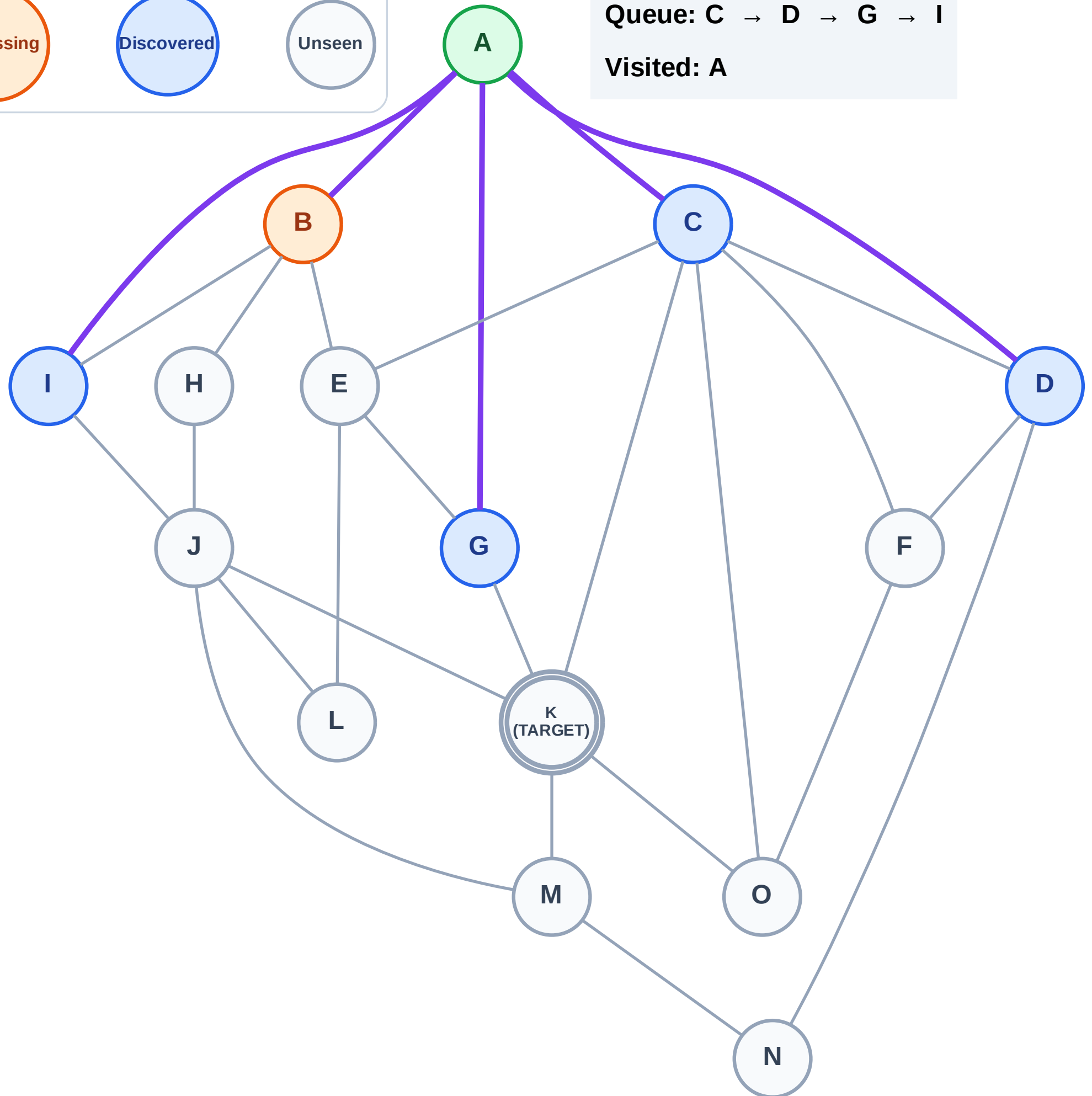
Step 4: Process node B

Legend



Queue: C → D → G → I

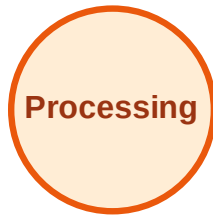
Visited: A



BFS Traversal

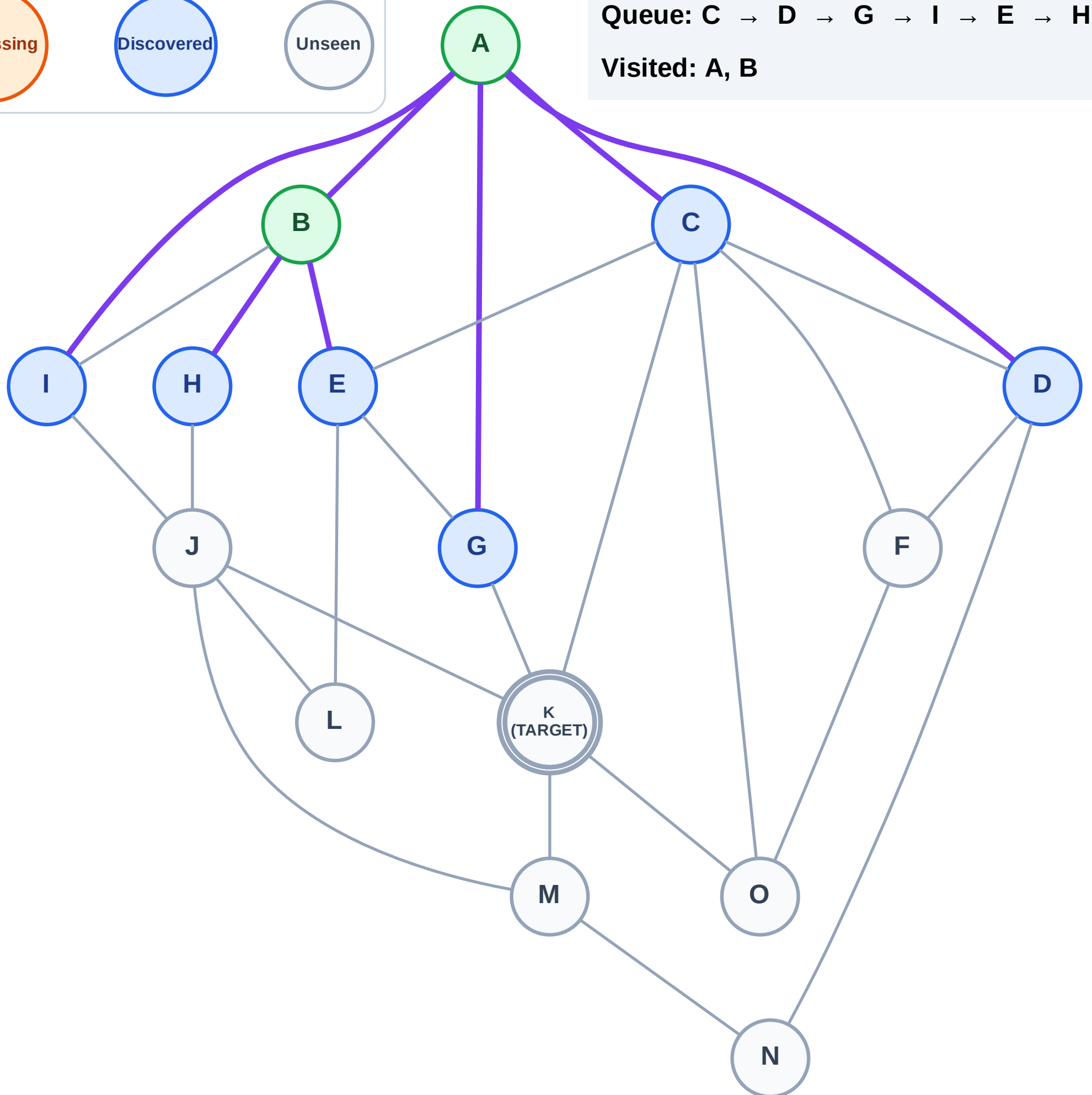
Step 5: Discover E, H from B

Legend



Queue: C → D → G → I → E → H

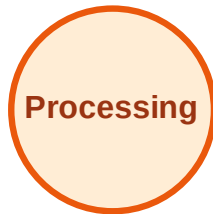
Visited: A, B



BFS Traversal

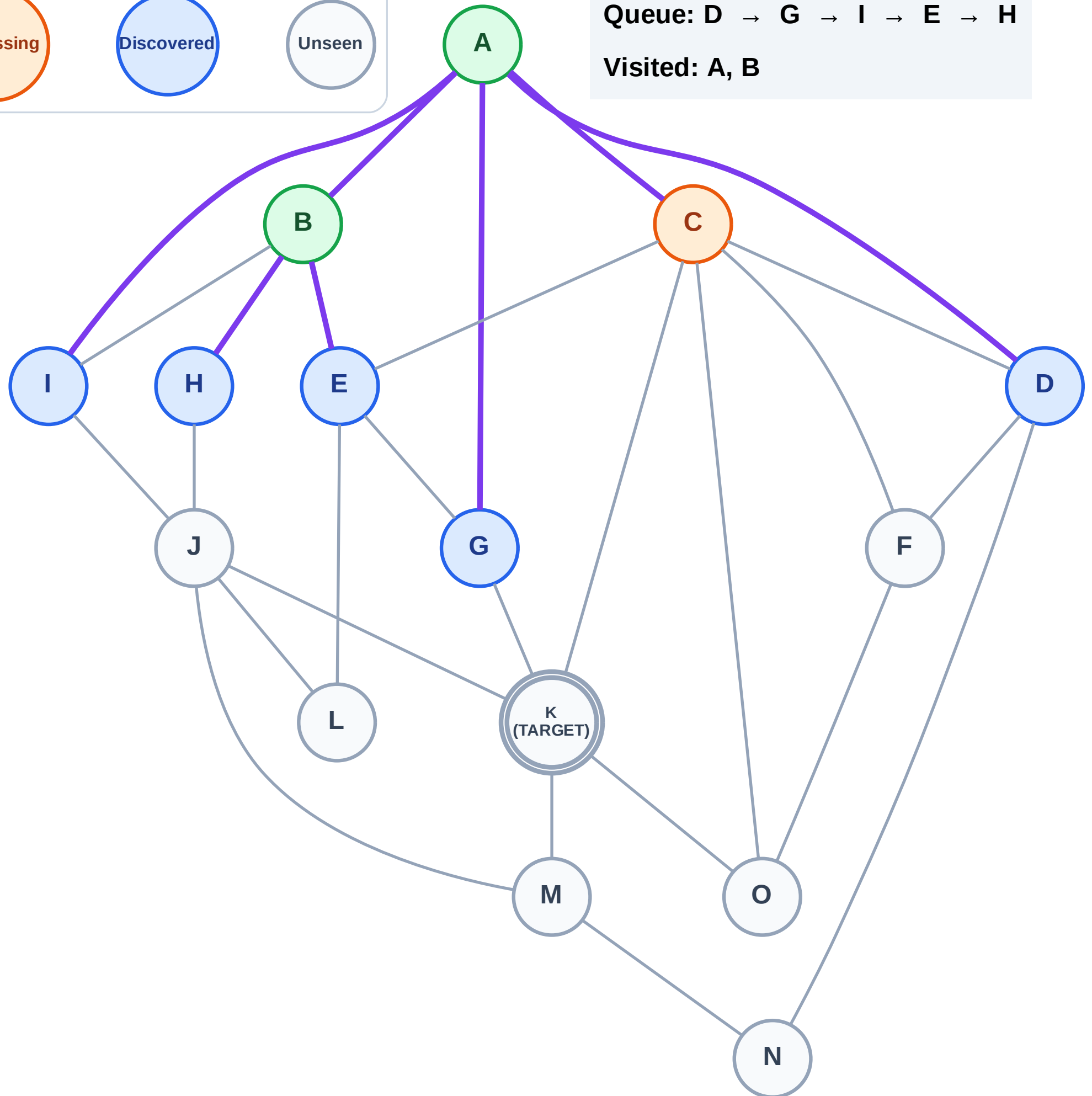
Step 6: Process node C

Legend



Queue: D → G → I → E → H

Visited: A, B



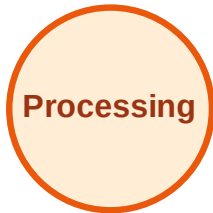
BFS Traversal

Step 7: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → C → K

Legend



Complete



Processing



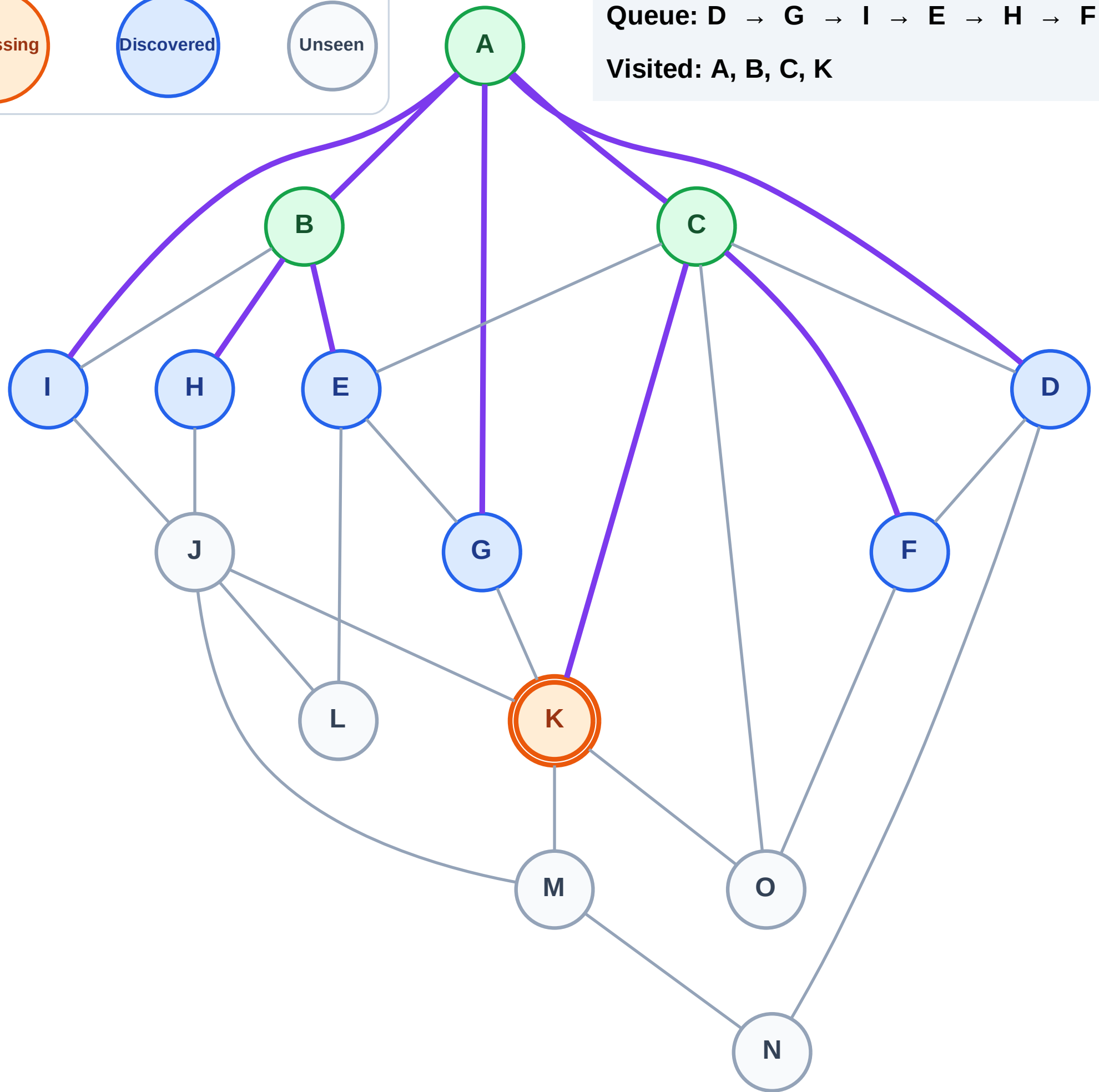
Discovered



Unseen

Queue: D → G → I → E → H → F

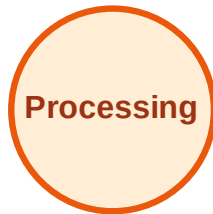
Visited: A, B, C, K



DFS Traversal

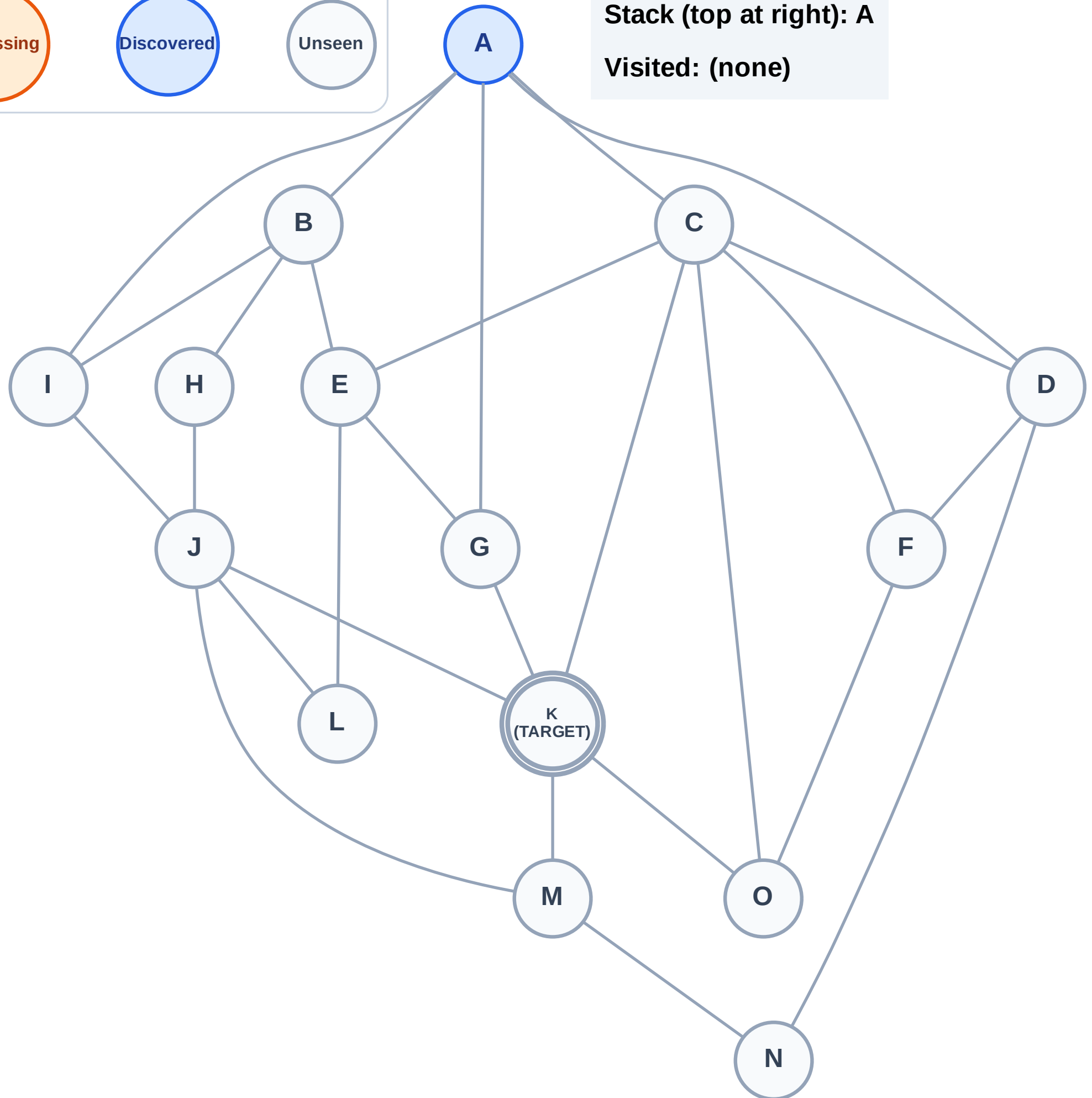
Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

Complete

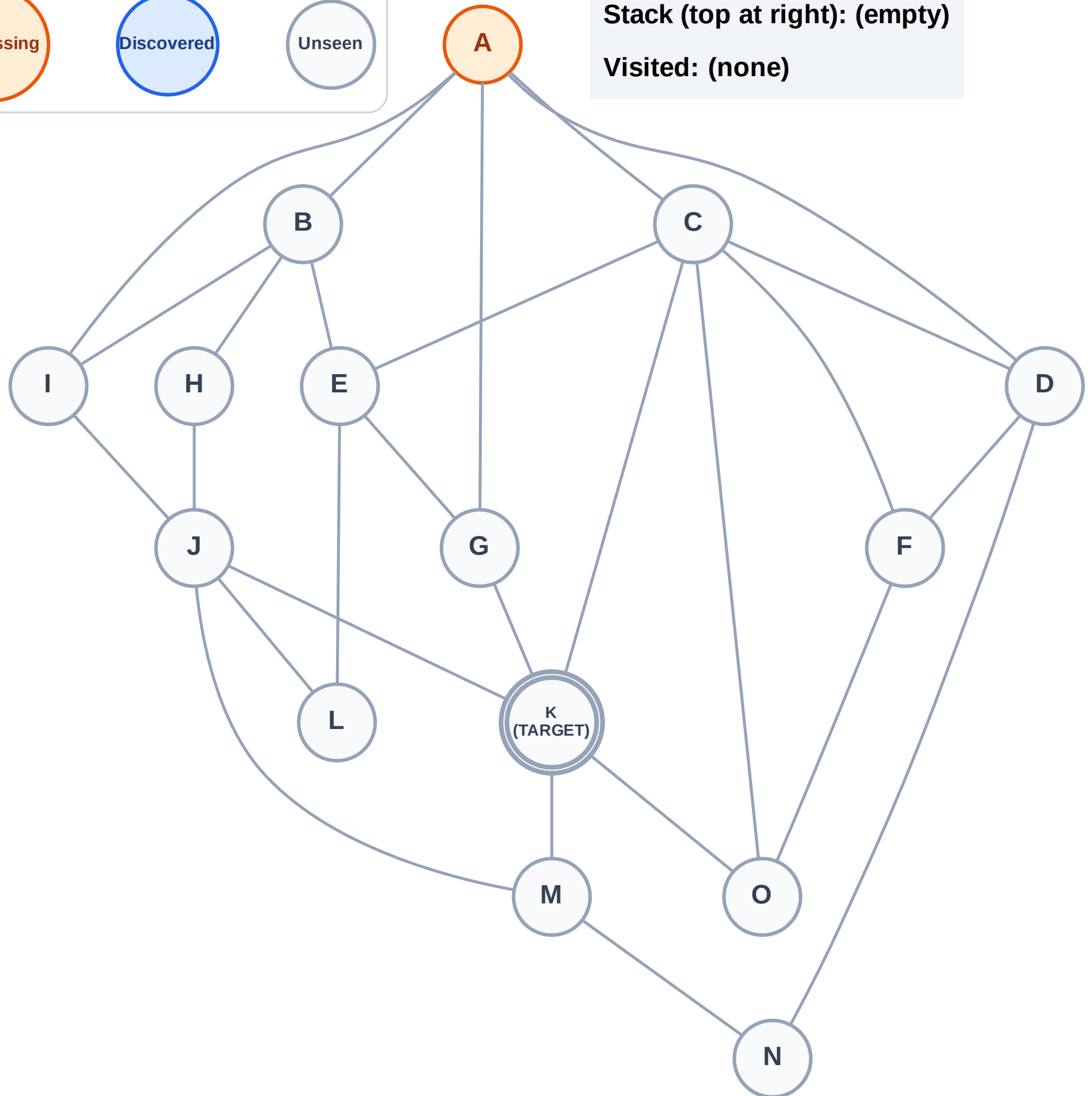
Processing

Discovered

Unseen

Stack (top at right): (empty)

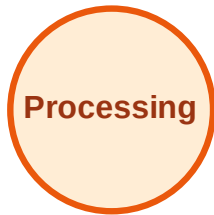
Visited: (none)



DFS Traversal

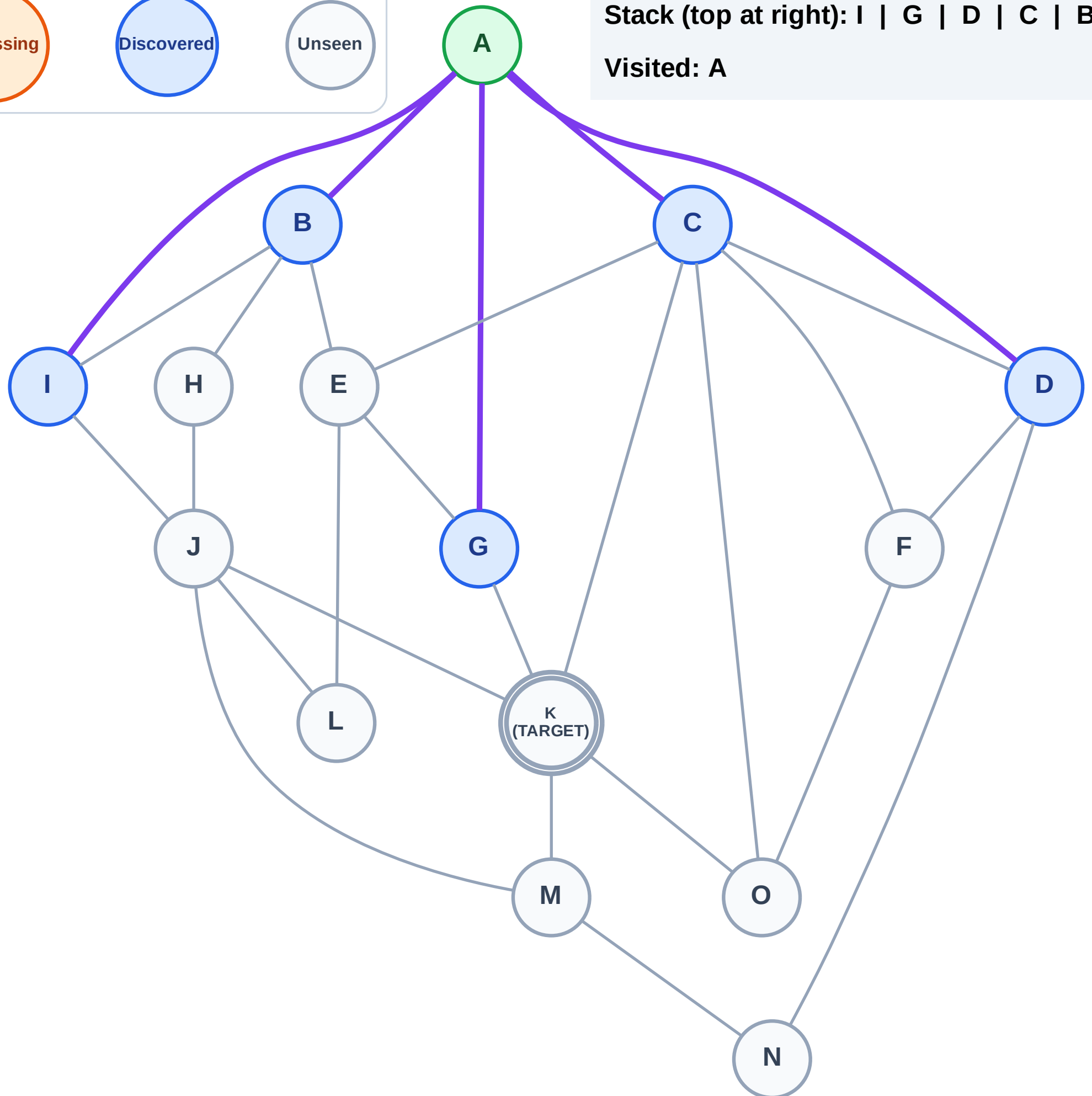
Step 3: Discover I, G, D, C, B from A

Legend



Stack (top at right): I | G | D | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

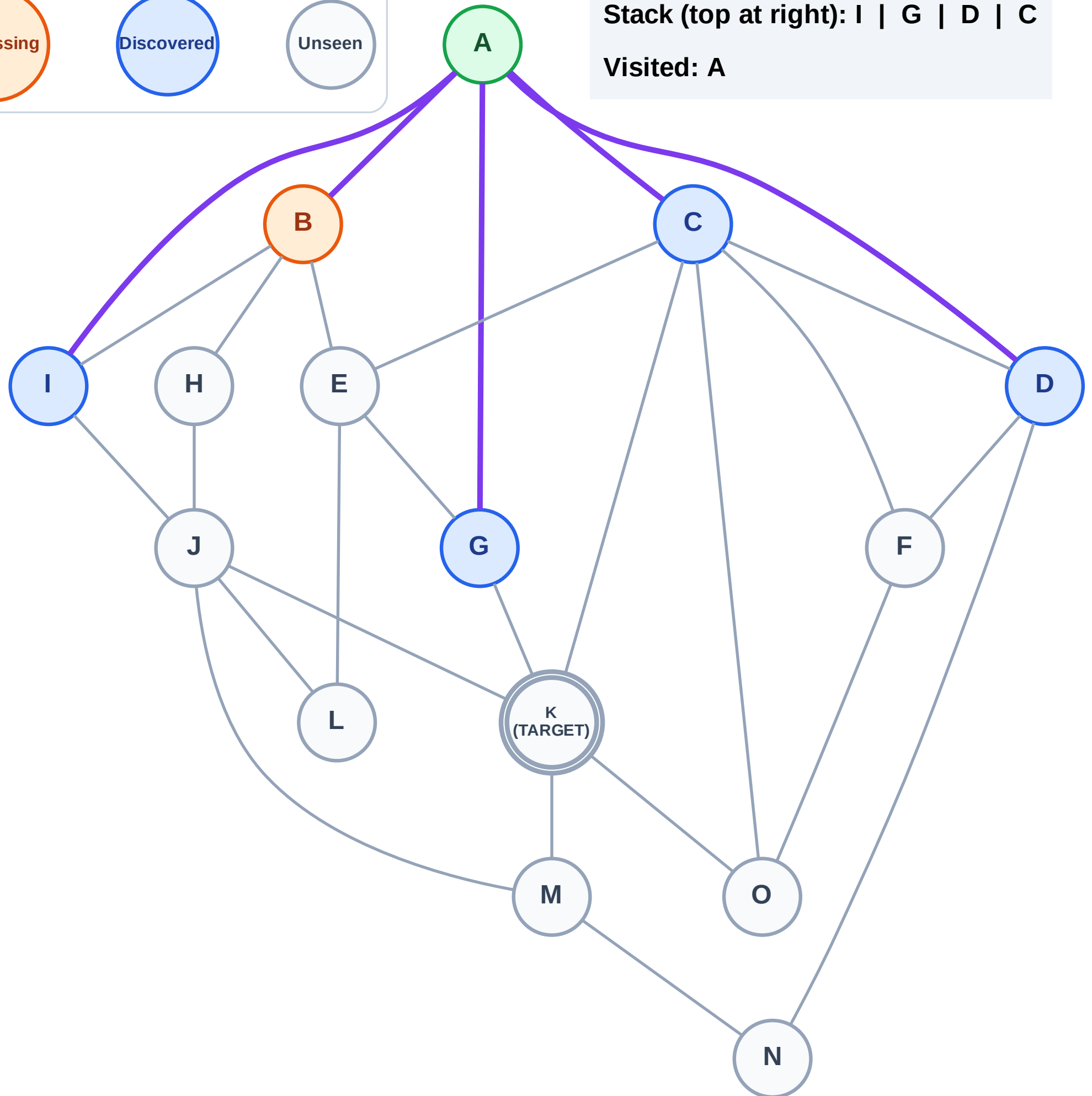
Processing

Discovered

Unseen

Stack (top at right): I | G | D | C

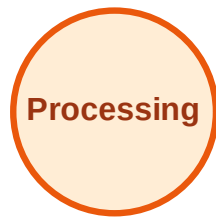
Visited: A



DFS Traversal

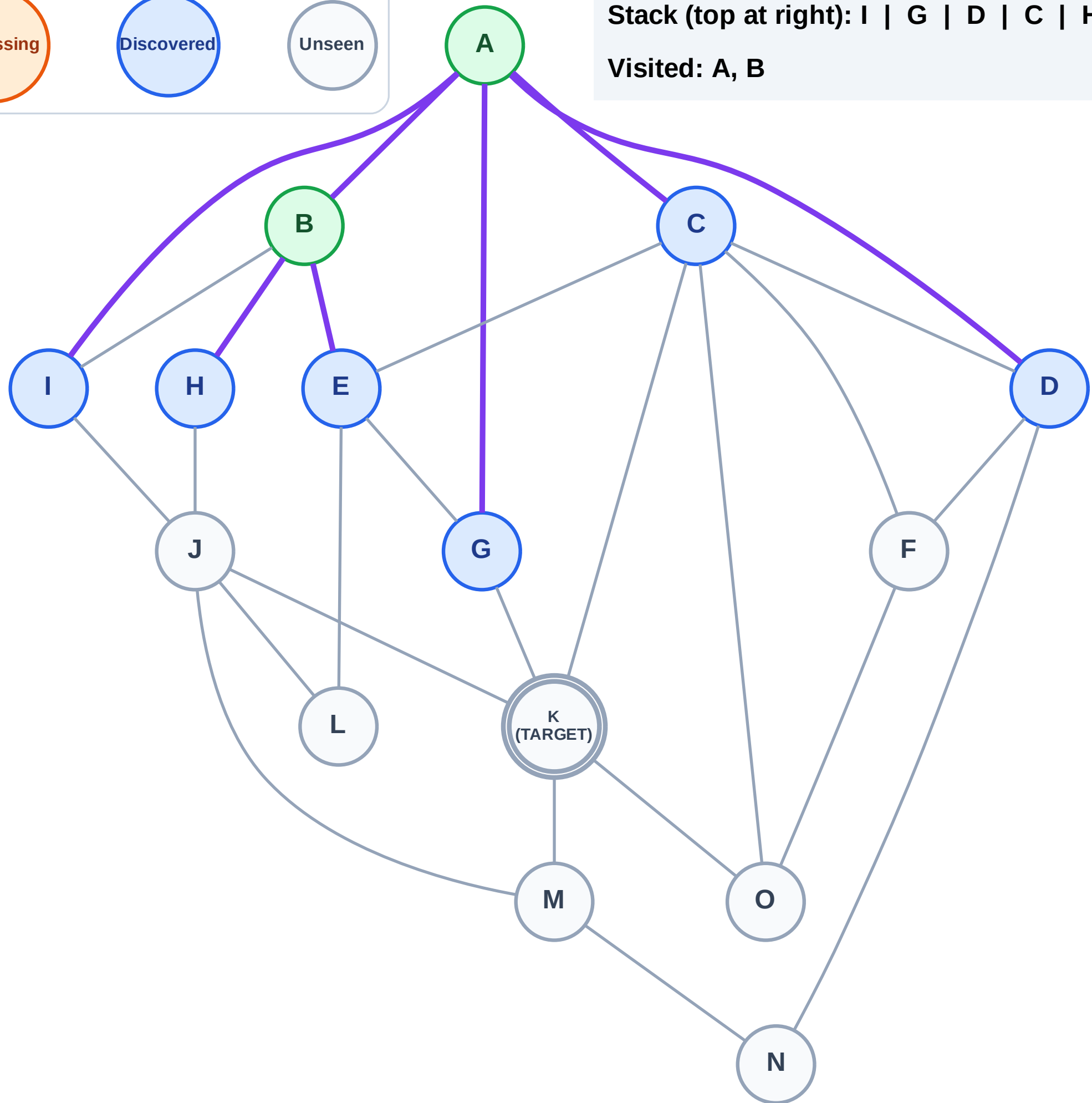
Step 5: Discover H, E from B

Legend

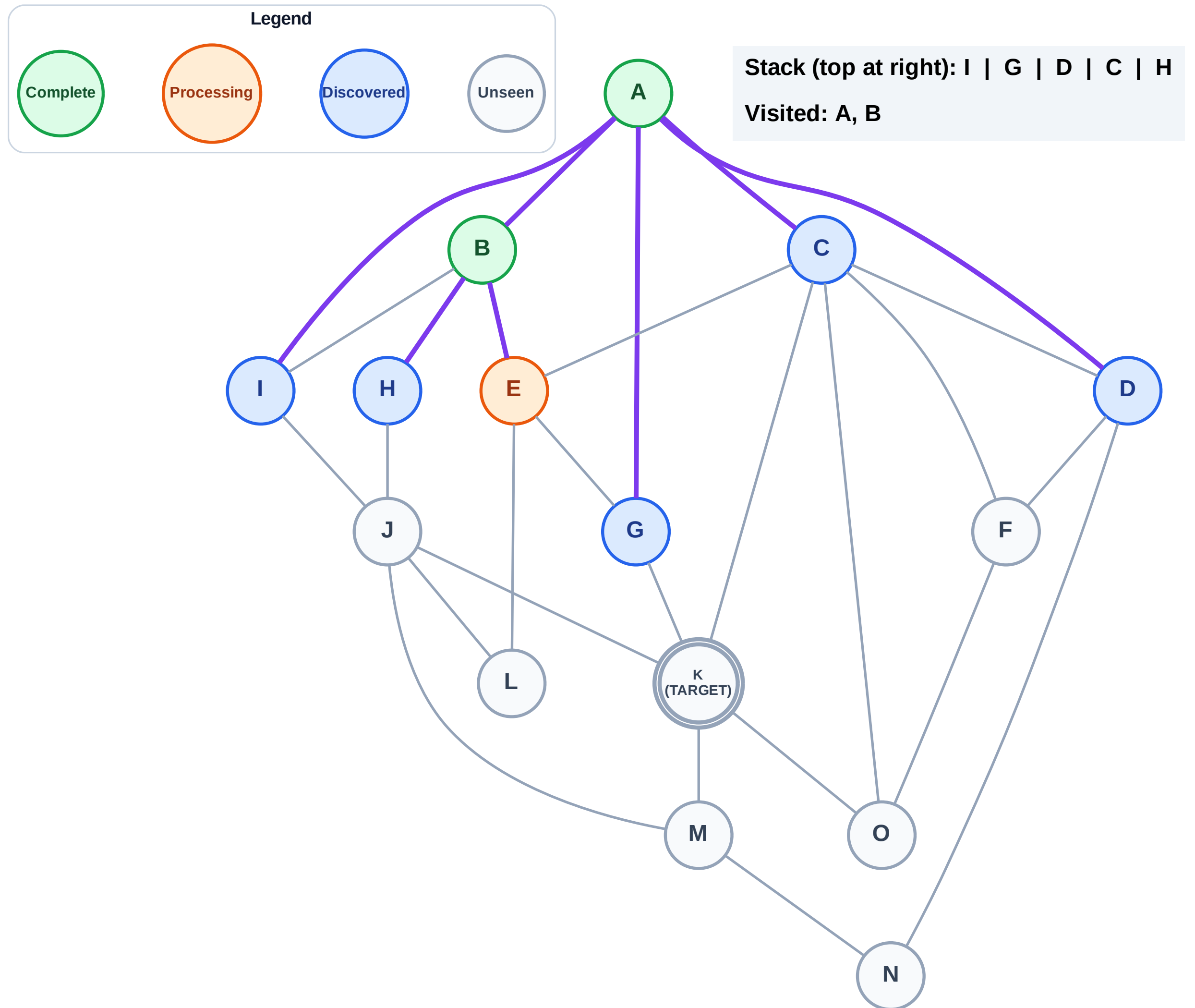


Stack (top at right): I | G | D | C | H | E

Visited: A, B



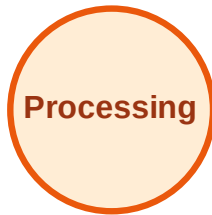
Step 6: Process node E



DFS Traversal

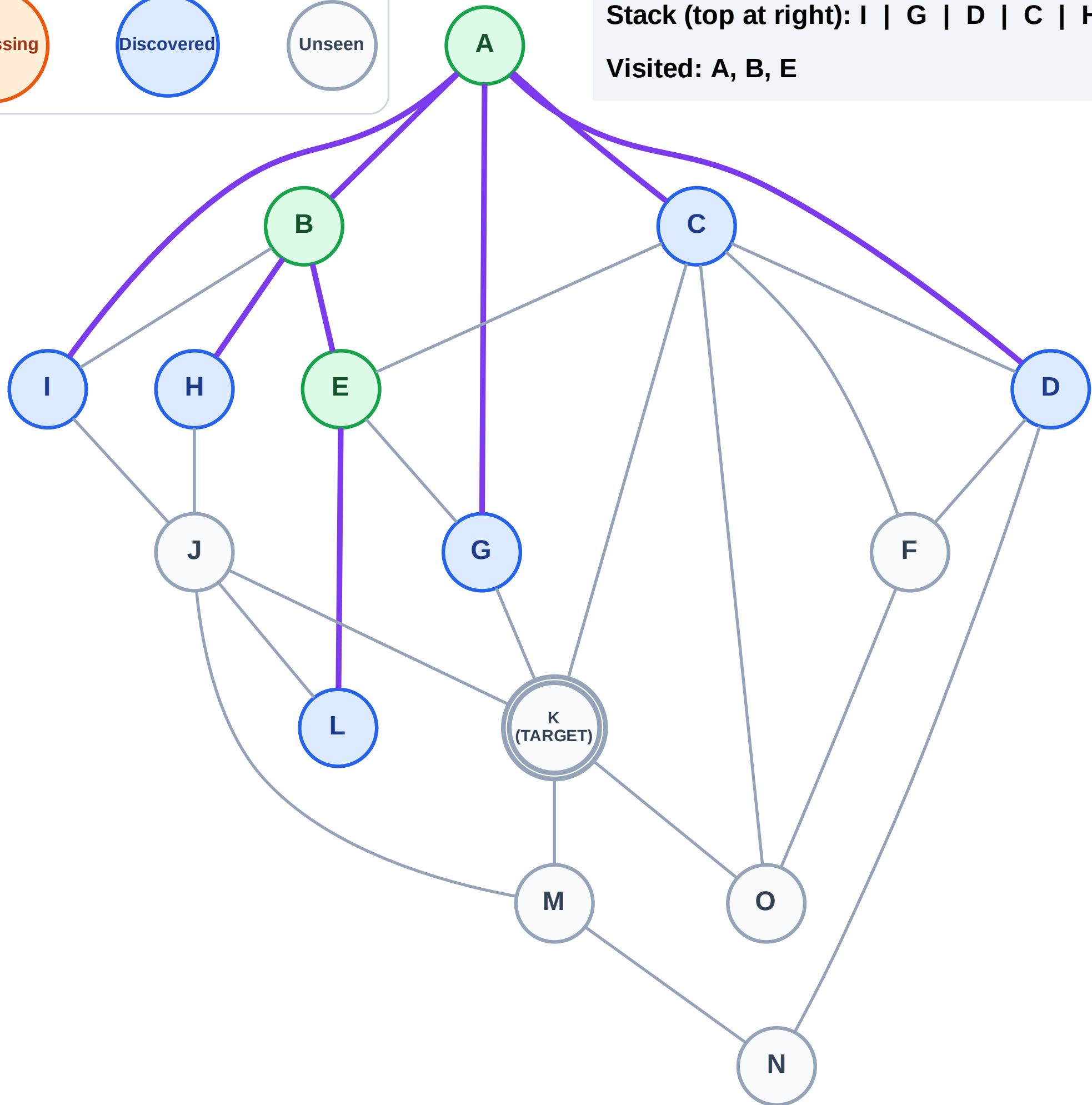
Step 7: Discover L from E

Legend



Stack (top at right): I | G | D | C | H | L

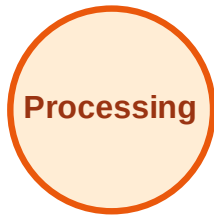
Visited: A, B, E



DFS Traversal

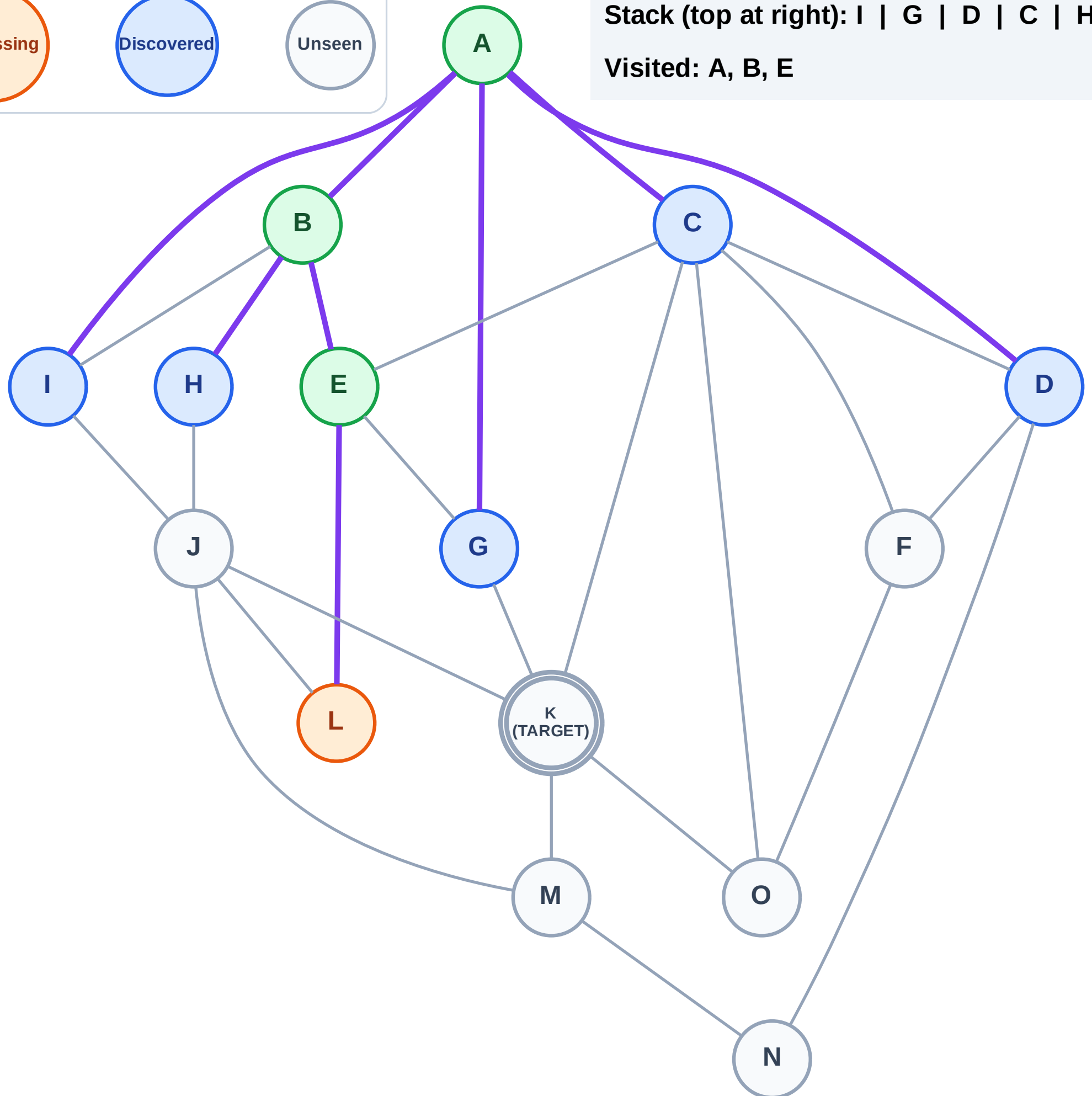
Step 8: Process node L

Legend



Stack (top at right): I | G | D | C | H

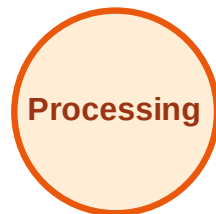
Visited: A, B, E



DFS Traversal

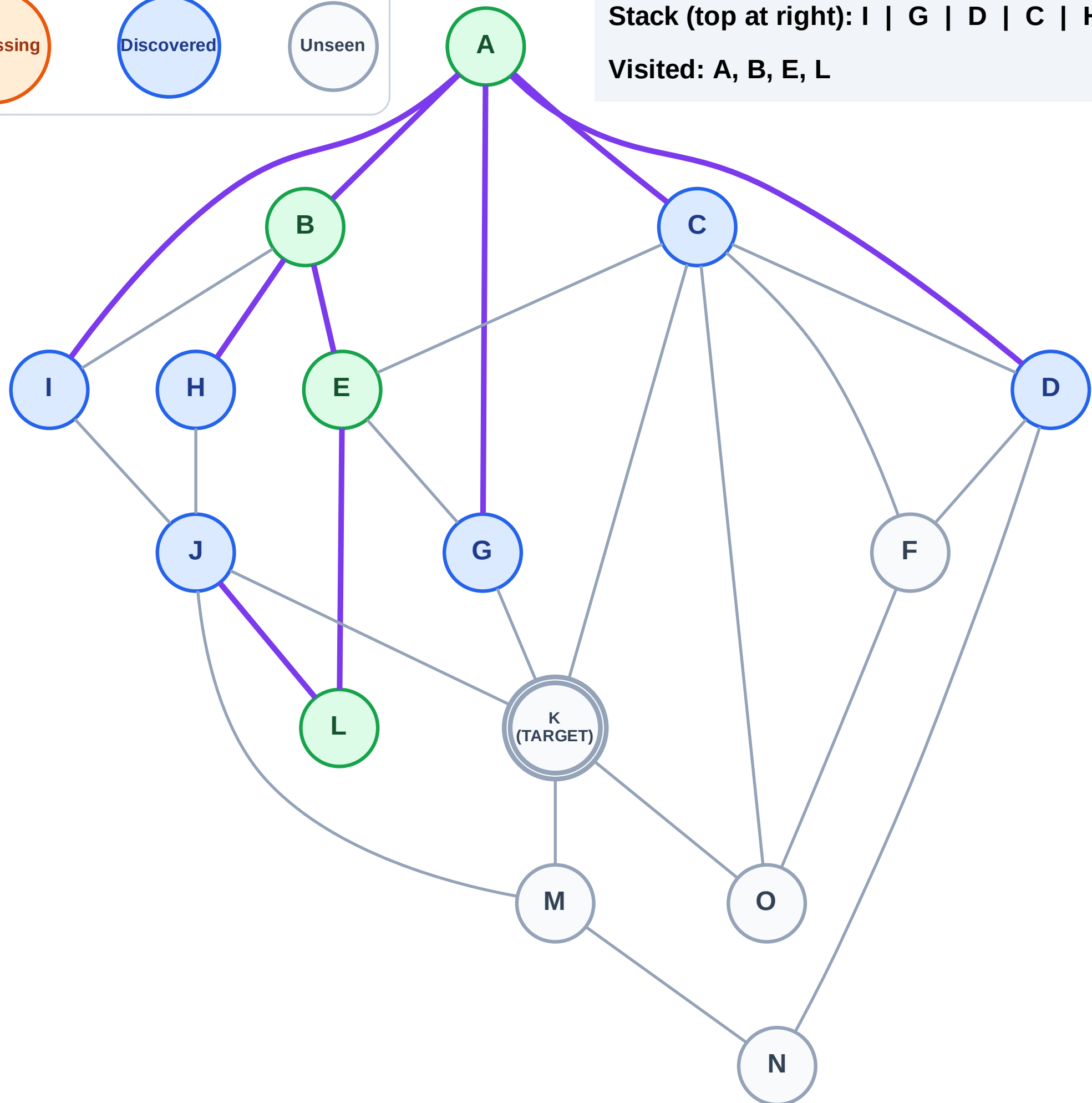
Step 9: Discover J from L

Legend



Stack (top at right): I | G | D | C | H | J

Visited: A, B, E, L



DFS Traversal

Step 10: Process node J

Legend

Complete

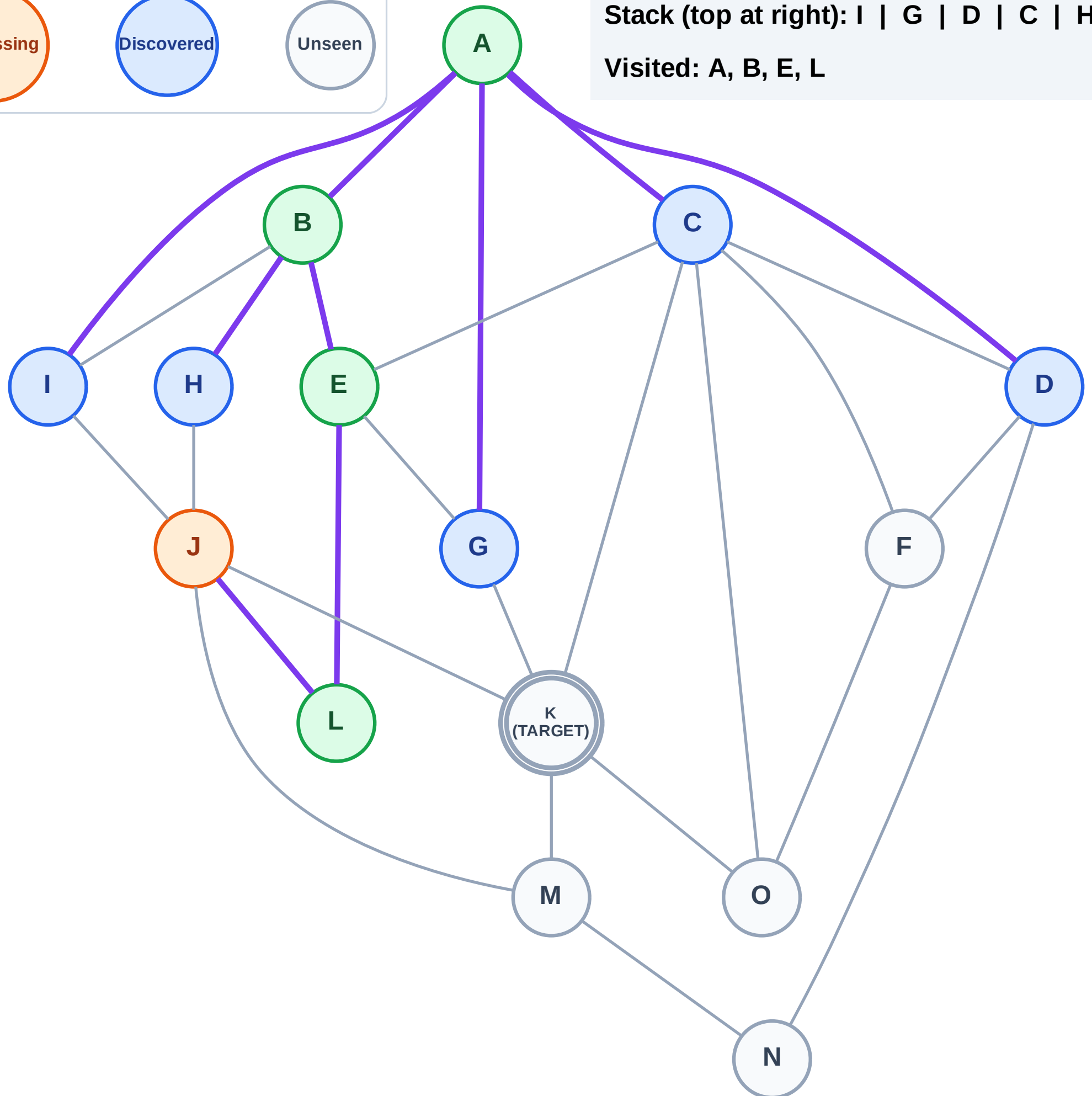
Processing

Discovered

Unseen

Stack (top at right): I | G | D | C | H

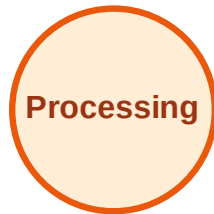
Visited: A, B, E, L



DFS Traversal

Step 11: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (5 hops): A → B → E → L → J → K

Legend



Stack (top at right): I | G | D | C | H | M

Visited: A, B, E, J, K, L

